THE COMPLETION OF AN AFFINE FORMAL ALGEBRAIC SPACE ALONG A CLOSED SUBSET OF ITS REDUCTION IS AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays the functor whose being an affine formal algebraic space is asserted; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 5 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Affine formal algebraic spaces as filtered colimits of affine schemes with thickenings as transition maps. Same chapter, Definition definition-affine-formal-algebraic-space, invoked at proof lines 7978-7980. Supplies the presentation whose completions are taken.
- The completion of an affine scheme along a closed subset is an affine formal algebraic space. Same chapter, Lemma lemma-completion-affine-is-affine-formal-algebraic-space, invoked at proof lines 7982-7985. Gives affineness of the completion of each stage.
- Morphisms between completions are base changes of the given morphisms, and reductions of completions agree. Same chapter, Lemma lemma-map-completions-representable at proof lines 7985-7989 and Lemma lemma-reduction-completion at proof lines 7989-7992. Verify the two conditions for the directed colimit statement.
- A directed colimit of affine formal algebraic spaces joined by closed immersions inducing isomorphisms on reductions is affine. Same chapter, Lemma lemma-colimit-affine-formal, invoked at proof lines 7992-7994. Gives the conclusion.

Source reference. Original lemma 0GVM: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 7868–7868. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIY: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3149–3149. Pointer outside the counted span, invoked once

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for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0APV: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3231–3231. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GB9: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3255–3255. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is an *affine formal algebraic space* if there exist

(1) a directed set Λ,
(2) a system $(X_\lambda, f_{\lambda\mu})$ over Λ in $(Sch/S)_{fppf}$ where
   (a) each X_λ is affine,
   (b) each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a thickening,

such that

$$X \cong \operatorname{colim}_{\lambda\in\Lambda} X_\lambda$$

as fppf sheaves and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of affine formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. Let X be an affine formal algebraic space over S. Let $T \subset |X_{red}|$ be a closed subset. Then the functor*

$$X_{/T} : (Sch/S)_{fppf} \longrightarrow Sets, \quad U \longmapsto \{f : U \to X : f(|U|) \subset T\}$$

is an affine formal algebraic space.

Proof. Write $X = \operatorname{colim} X_\lambda$ as in Definition 0.1. Then $X_{\lambda,red} = X_{red}$ and we may and do view T as a closed subset of $|X_\lambda| = |X_{\lambda,red}|$. By Lemma 0AIY for each λ the completion $(X_\lambda)_{/T}$ is an affine formal algebraic space. The transition morphisms $(X_\lambda)_{/T} \to (X_\mu)_{/T}$ are closed immersions as base changes of the transition morphisms $X_\lambda \to X_\mu$, see Lemma 0APV. Also the morphisms $((X_\lambda)_{/T})_{red} \to ((X_\mu)_{/T})_{red}$ are isomorphisms by Lemma 0GB9. Since $X_{/T} = \operatorname{colim}(X_\lambda)_{/T}$ we conclude by Lemma 0GVM. □